

Summary

Data Engineer II at Amazon with 5+ years of experience building large-scale data lake infrastructure, event-driven pipelines, and AI-powered automation systems. Designed and owned a 63+ pipeline medallion architecture data lake serving global operations, built AI agents on Amazon Bedrock that automate metadata enrichment and pipeline triage, and led cross-org platform migrations at scale. Deep expertise in AWS-native data infrastructure, PySpark, and distributed systems.

Education

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| <b>Master of Science in Computer Engineering</b><br>George Mason University, Fairfax, Virginia<br><i>Outstanding Academic Achievement Award</i> | <b>May 2021</b> |
| <b>Bachelor of Engineering in Electronics and Communication Engineering</b><br>Osmania University, Hyderabad, India                             | <b>May 2019</b> |

Technical Proficiencies

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|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Languages:          | Python, SQL, PySpark, Scala                                                                                                                               |
| Cloud Technologies: | Azure, AWS, Databricks                                                                                                                                    |
| Databases:          | Redshift, Aurora MySQL, DynamoDB, DocumentDB, Snowflake, S3, Apache Iceberg                                                                               |
| AWS Services:       | Glue, EMR Serverless, Lambda, Athena, Lake Formation, DataZone, QuickSight, EventBridge, SNS, SQS, SES, CloudFront, API Gateway, CodePipeline, CloudWatch |
| Azure Services:     | Databricks, Storage accounts, Key vaults, App Services, Virtual networks                                                                                  |
| AI & Agents         | Amazon Bedrock, Bedrock AgentCore, Strands Agents SDK, Langfuse                                                                                           |
| Tools:              | Jupyter Notebook, Git, Jira, Visual Studio Code                                                                                                           |

Professional Experience

Amazon

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| <b>Data Engineer II - Austin, TX</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>May 2024 – Present</b>  |
| <ul style="list-style-type: none"><li>Owned the enterprise data lake platform powering 63+ production pipelines, built on a 4-layer medallion architecture (Landing → Raw → Transform → Refine) with automatic compute selection across Lambda, AWS Glue, and EMR Serverless, and Apache Iceberg for ACID compliance on high-volume datasets.</li><li>Automated registration and AI enrichment of 6,000+ datasets into the enterprise data catalog, replacing a 30-min-per-table manual process with a Strands Agent on Bedrock AgentCore that auto-generates metadata, descriptions, and data classifications at scale.</li><li>Built an AI agent deployed on an internal agent platform that triggers on pipeline failure tickets, analyzes root cause using domain knowledge, and posts a detailed diagnosis comment directly on the ticket, reducing mean time to resolution from 4-5 days to 1-2 days.</li><li>Reduced pipeline onboarding time from half a day to 30 minutes with a conversational AI interface where teams configure and launch new data sources through chat. Replaced a fixed weekly release cycle with instant production deployments, enabling same-day pipeline launches.</li><li>Led org-wide migration to a next-generation data catalog platform, owning documentation and cross-team coordination to ensure zero disruption across the Engineering &amp; Analytics organization.</li><li>Automated analytics access provisioning by building a system that syncs identity groups to QuickSight permissions in near real-time, coordinating across LDAP, IAM, and Lake Formation to grant analysts instant access to sensitive datasets when authorized.</li></ul> |                            |
| <b>Data Engineer - Seattle, WA</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | <b>Apr 2022 – Apr 2024</b> |
| <ul style="list-style-type: none"><li>Eliminated ~200 daily manual email reviews by building an event-driven ticket routing system that auto-routes warehouse sensor failures, deduplicates multi-device alerts, and auto-resolves transient issues.</li><li>Built multi-vendor fleet telemetry pipelines ingesting data from 40+ API endpoints across US and European fulfillment centers, completing a 2-year historical backfill into the enterprise data lake while retiring a legacy Redshift cluster.</li><li>Expanded the data lake to handle sensitive and restricted datasets by completing a formal security certification process, authoring architecture documentation and threat mitigations through cross-functional review.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                            |

- Reduced an executive analytics pipeline runtime by 98% (from 40+ hours to 30 minutes) by identifying 990GB of unnecessary data processing and redesigning with partition pruning and targeted S3 crawlers, unblocking weekly reporting for multiple business units.
- Migrated 600+ users across Europe off 9 legacy tools to a unified data platform in a planned one-day cutover with zero data loss, owning schema mapping for 50+ tables and all downstream compatibility pipelines.

## **Blue Cloud**

**Associate Data Engineer - Rolling Meadows, IL**

**Jun 2021 – Apr 2022**

- Built PySpark data pipelines in Azure Databricks for ETL from diverse file formats into Azure Data Lake Storage and Amazon S3, providing insights into customer usage patterns.
- Developed and deployed AWS Lambda functions in Python for data extraction, parsing, and ingestion of nested JSON from S3 into Snowflake.
- Engineered Snowflake Snowpipe for real-time data loading with internal and external stages, and defined roles, privileges, and virtual warehouse sizing for various workload types.
- Implemented databases, schemas, and tables in Databricks and Snowflake for diverse data handling and automated pipeline maintenance, improving ETL efficiency across multiple workloads.
- Provided L2/L3 production support for 100+ clients on the JLL Azara platform and contributed to integration infrastructure design across multiple teams.

## **George Mason University**

**Graduate Teaching Assistant - Fairfax, VA**

**Jan 2020 – May 2021**

- Assisted Computer Science Department in Graduate level courses – ‘The Essentials of Computer Science’ and ‘Big data technologies’.
- Assisted the Information Sciences and Technology Department in Undergraduate level courses – ‘Security Accreditation of Information Systems’ and ‘Information Security Principles’
- Collaborated with colleagues and professors to discuss student progress and helped develop skills during weekly office hours.

## **Certifications**

AWS Certified: AI Practitioner | 2026

AWS Certified: Cloud Practitioner | 2023

Microsoft Certified: Azure Data Fundamentals | 2021

Microsoft Certified: Azure Data Engineer Associate | 2021

Snowflake Certified: SnowPro Core | 2021